

EXAM 7 – SPRING 2015

22. (2.25 points)

An actuary is modeling the variability of an insurer's homeowners ultimate losses for the next accident year as part of an internal solvency model.

a. (1.5 points)

Identify and briefly explain three key elements of uncertainty inherent to the loss modeling process.

b. (0.75 point)

The actuary uses a normal distribution to model the losses, with the parameters selected using five years of internal loss history. The actuary has just learned of a recent court ruling affecting homeowners coverage triggers in the state where the insurer writes the most premium.

For each of the three elements of uncertainty identified in part a. above, suggest an improvement to the actuary's modeling process that would decrease the overall uncertainty in modeled losses.

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QUESTION 22	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: C2: Describe the use of enterprise-wide risk modeling and aggregation techniques.
SAMPLE ANSWERS	
Part a: 1.5 points	
<i>Sample Answer 1</i> Model Risk – models can't fully capture the complexity of the insurance process. Estimation Risk – historical data may not be enough to accurately estimate model parameters. Projection Risk – parameter estimates may not be equal to historical estimates for projection period due to trend / inflation. <i>Sample Answer 2</i> Model Risk – risk that models can't accurately describe insurance process / data Parameter Risk – risk associated with identifying and estimating predictors of claims Structural Risk – risk that parameters vary with time. <i>Sample Answer 3</i> Specification Risk – risk that the model cannot accurately model the insurance process. Parameter Selection Risk – risk that not all parameters / trends can be properly identified. Data Error – risk that the data is not credible or the person analyzing the data does not fully understand it. <i>Sample Answer 4</i> Model Selection – whether the model selected accurately reflects the insurance losses given that it is based on a sample. Parameter Selection – whether the parameters selected are appropriate to match the real world experience Extreme Events – events will impact the uncertainty of the modeling process. Ex: catastrophes	
Part b: 0.75 point	
<i>Sample Answer 1</i> Use distribution with heavier tails to model extreme loss nature of cat exposed HO losses Use more years of historical data / industry data to increase credibility of data used for parameter selection Adjust parameter estimates using judgment to reflect effect of coverage trigger on loss distribution for projected period. <i>Sample Answer 2</i> Model Risk – normal may not reflect skewness of insurance process – could use lognormal Parameter Risk – could look at longer historical period of internal data or combine with industry data. Structural Risk – due to court ruling, historical parameters need adjusting (at least for largest state)	

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Sample Answer 3

Specification – Use a lognormal distribution to model losses. This is asymmetric and better fits insurance losses than normal.

Parameter selection error – incorporate the effect of the court ruling into a parameter in the model to increase accuracy of expected future losses.

Data Error – Use ten years of internal loss experience instead of five to increase credibility.

Sample Answer 4

Model Selection: Normal distribution might not be the best. It is symmetric and doesn't have fat tails. Could consider using lognormal which is better at capturing losses at higher probabilities.

Parameter Selection – Use a longer history than five years, or use external loss data as well to reduce uncertainty about parameters.

Extreme Events – recent court ruling could be considered an extreme event. Actuary should adjust past data for the new coverage trigger so that the model is accurate going forward.

EXAMINERS' REPORT

Candidates were expected to know some basic concepts in ERM models and specifically the loss modeling process. Candidates generally performed well.

Part a

Most candidates were able to identify and describe three key elements of uncertainty inherent in the loss modeling process. While these show up on the syllabus with various names, any valid identification was accepted for full credit. For example, Model Risk, Model Error, and Specification Error each identify one of the key elements. Where a candidate would not have received full credit was using a simple restatement of the identification as the brief description of the key element. Several candidates mentioned process risk or volatility, which is a key part of insurance risk but is not a key element of uncertainty in the loss modeling process.

Part b

This part of the question required candidates to apply a simple scenario to the key elements of uncertainty from part a. The three improvements from the scenario were to use a distribution other than normal (model risk), use more or less years of data depending on volume or use external data (parameter risk), and to adjust losses for the court ruling (structural risk). Even where candidates erred in part a, many were able to provide improvements in part b that earned credit.